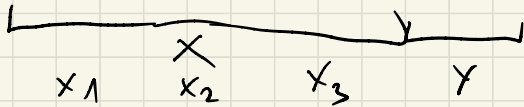


BOJA	TIP	POREKLO	UKRAĐEN?
CRVENA	SPORTSKI	DOMAĆI	DA
CRVENA	SPORTSKI	DOMAĆI	Ne
CRVENA	SPORTSKI	DOMAĆI	DA
ŽUTA	SPORTSKI	DOMAĆI	Ne
ŽUTA	SPORTSKI	UVEZEN	DA
ŽUTA	SUV	UVEZEN	Ne
ŽUTA	SUV	UVEZEN	DA
ŽUTA	SUV	DOMAĆI	Ne
CRVENA	SUV	UVEZEN	Ne
CRVENA	SPORTSKI	UVEZEN	DA



Novi:

CRVENA	SUV	DOMAĆI	?
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$$A: \prod_i P(x_i | Y_{DA}) \cdot P(Y_{DA})$$

$$B: \prod_i P(x_i | Y_{Ne}) \cdot P(Y_{Ne})$$

$$A > B \Rightarrow \boxed{DA} \quad B > A \Rightarrow \boxed{Ne}$$

$$P(Y) = \frac{\text{COUNT}(Y)}{N}$$

$$P(Y|X) = \frac{\text{COUNT}(X,Y)}{\text{COUNT}(X)} \cdot \frac{1}{N} = \frac{P(X,Y)}{P(X)}$$

$$P(Y|X_1, X_2, X_3) ?$$

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

$P(X_1|Y) \perp P(X_2|Y) \checkmark$
 $P(X_1) \perp P(X_2) \times$

$$P(X|Y) \neq P(X_1|Y) \cdot P(X_2|Y) \cdot P(X_3|Y)$$

$$P(Y|X) = \frac{\prod_i P(x_i | Y) \cdot P(Y)}{P(X)} \sim \boxed{\prod_i P(x_i | Y) \cdot P(Y)}$$

MODEL

$$\hat{Y} = \underset{Y}{\text{ARGMAX}} \prod_i P(x_i | Y) \cdot P(Y)$$

PREDICT

BOJA	TIP	POREKLO	UKRADEN?
CRVENA	SPORTSKI	DOMAĆI	DA
CRVENA	SPORTSKI	DOMAĆI	Ne
CRVENA	SPORTSKI	DOMAĆI	DA
ŽUTA	SPORTSKI	DOMAĆI	Ne
ŽUTA	SPORTSKI	UVEZEN	DA
ŽUTA	SUV	UVEZEN	Ne
ŽUTA	SUV	UVEZEN	DA
ŽUTA	SUV	DOMAĆI	Ne
CRVENA	SUV	UVEZEN	Ne
CRVENA	SPORTSKI	UVEZEN	DA

MODEL:

$P(Y)$		$P(Y)$	
		DA	0,5
		Ne	0,5
$P(BOJA Y)$		C	$\bar{C}$
		DA	3 2 5
		Ne	2 3 5
$P(TIP Y)$		S	SUV
		DA	4 1
		Ne	2 3
$P(POREKLO Y)$		D	U
		DA	2 3
		Ne	3 2

MODEL

LEARN

μ_{DA}, σ_{DA}

μ_{Ne}, σ_{Ne}

PREDICT

Novi:

CRVENA	SUV	DOMAĆI	? NE
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0,75

$$P(DA|C, SUV, D) \sim 0,6 \cdot 0,2 \cdot 0,4 \cdot 0,5 = 0,024$$

$$P(Ne|C, SUV, D) \sim 0,4 \cdot 0,6 \cdot 0,6 \cdot 0,5 = 0,072 \quad \boxed{Ne}$$

$$P(X) = \sum_j P(X|Y_j) \cdot P(Y_j) = \sum_j P(X, Y_j)$$

$$P(X) = \frac{0,024 + 0,072}{0,096}$$

$$P(Y_{Ne}|X) = 0,75$$

$0,2 \cdot 0,002 \cdot 0,00025 \cdot \dots \Rightarrow 0$

UNDERFLOW

$$45 \cdot 10^{-40}$$

$$\underbrace{A \cdot B} \quad \log(A \cdot B) = \log A + \log B$$

$\prod \rightarrow \Sigma$

$$\log(\quad) + \log(\quad) + \log(\quad)$$

$$\odot \log(A \cdot B) = A \cdot B$$

$$\odot \underbrace{(\log A + \log B)} = \underbrace{A \cdot B}$$

Age				
20				
25				
28				
45				
48				

μ_{DA}, σ_{DA}
 μ_{He}, σ_{He}

	20-21	22	23	...
DA				
He				

$P(X_i | Y)$

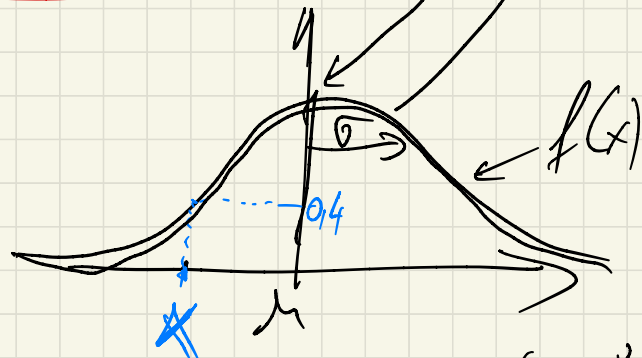
SCIPY. STATS. NORM. PDF

$$\text{PDF}(x, \mu, \sigma) \rightarrow 0.74$$

25

$P(X|Y)$ (NUM)
 $P(X|Y)$ (MODEL)

$$P(X|Y) \sim \mathcal{N}(\mu, \sigma)$$



$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$